Watershed *Segmentation.*

A topographic method for partitioning images — where pixels become terrain and water carves the boundaries.

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What is image *segmentation*?

Splitting an image into regions that mean something — a tumour, a cell, a building, a road. Almost every computer vision task starts with this question: what is where?

DOMAIN · 01

Medical imaging

Delineating tumour boundaries in CT and MRI scans, separating cells in histology slides.

DOMAIN · 02

Remote sensing

Identifying land cover, water bodies, and built environment in satellite imagery.

DOMAIN · 03

Industrial · vision

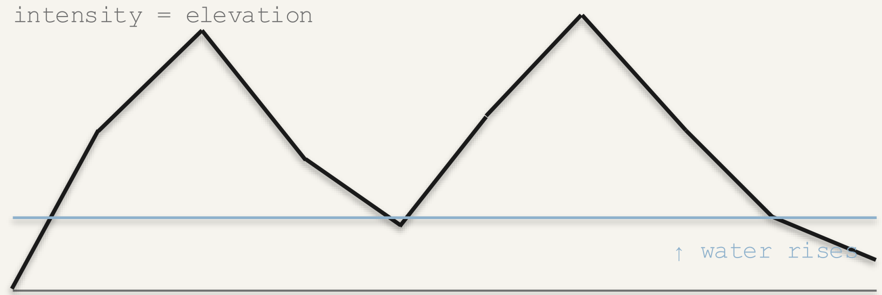
Particle sizing in fluid flow; instance segmentation for autonomous driving.

A greyscale image *is* a terrain.

Bright pixels = mountains. Dark pixels = valleys.

If it rained on this landscape, where would the water go?

SCHEMATIC



1D cross-section of an image as terrain — simplified.

Words you'll hear *throughout* this talk.

01 / VALLEY FLOOR

Regional minimum

A flat patch in the landscape where every step you take outward goes uphill — the bottom of a valley.

02 / DRAINAGE REGION

Catchment basin

All the pixels whose water would flow downhill into the same valley. One basin per valley.

03 / THE BOUNDARY

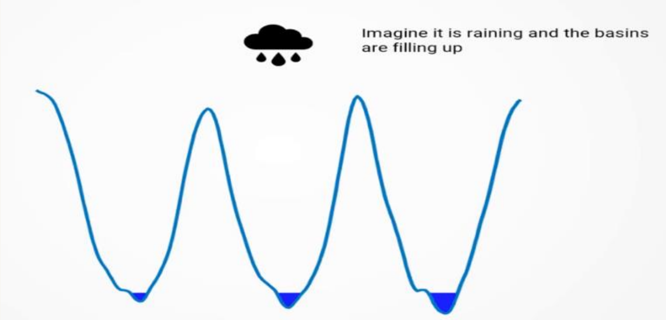
Watershed line

The ridge between two valleys — where water could fall either way. Exactly where we want our segmentation boundary.

STEP BY STEP

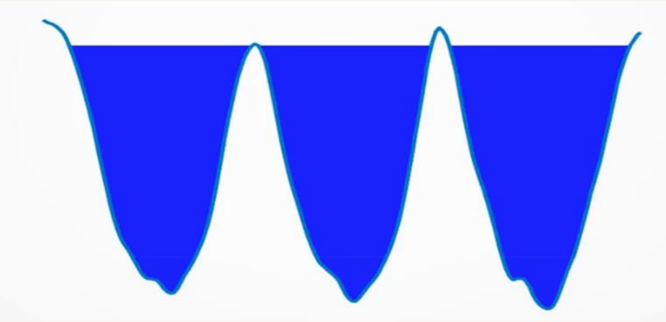
Submerge the terrain from *below*.

STAGE 01 · H IS LOW



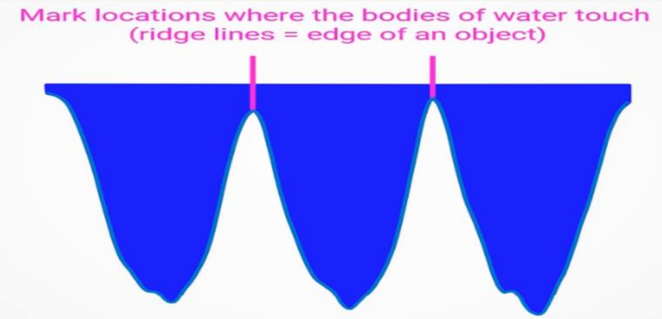
Deepest minima flood first — isolated lakes appear.

STAGE 02 · H RISING



Two lakes are about to touch.

STAGE 03 · DAM BUILT



Dams are built where neighbouring basins would touch becoming the watershed boundaries

flooded at level h = { every pixel with intensity ≤ h }

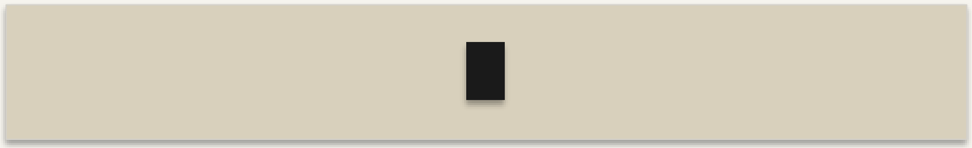
Not just a metaphor — this is a constructive definition that maps onto an efficient flooding algorithm.

THE OBVIOUS APPROACH FAILS

Watershed on the image directly? *Almost never*.

F · RAW INTENSITY

Run watershed on f

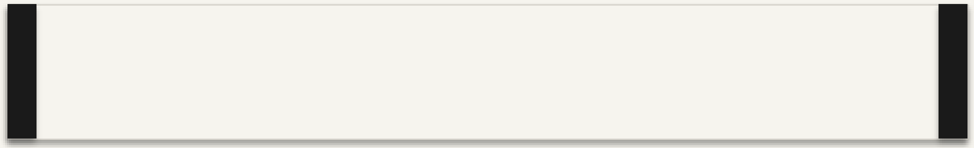


darkest pixel

Basins seed inside the object (darkest pixel) — boundaries land in the wrong place.

G(F) · GRADIENT MAGNITUDE

Run watershed on g(f)



edges = peaks of g(f)

Basins seed in flat interiors; ridges fall exactly on the edges.

WHAT WE RUN WATERSHED ON

The *morphological* gradient.

$$g(f) = (f \oplus B) - (f \ominus B)$$

DILATION

$(f \oplus B)(x) = \max\{ f(y) : y \in B_x \}$

Maximum intensity in the local neighbourhood B around x.

EROSION

$(f \ominus B)(x) = \min\{ f(y) : y \in B_x \}$

Minimum intensity in that same neighbourhood.

THEIR DIFFERENCE

Local intensity spread

Zero on flat regions. Large wherever intensity changes rapidly — i.e. at edges.

B is a small structuring element (typically 3×3). Smaller B → finer edges; larger B → coarser boundaries. It is a scale knob.

THE ALGORITHM · LINEAR-TIME WITH ORDERED QUEUES

Pop pixels low-to-high. *Three cases*. That's it.

CASE 01

No labelled neighbours yet.

→ Seed a new basin B_k.

A fresh regional minimum has been found.

CASE 02

All labelled neighbours belong to the same basin.

→ Assign p to that basin.

The pixel is interior to a region.

CASE 03

Labelled neighbours come from two or more basins.

→ Mark p as WSHED.

The dam between two merging lakes.

Repeat until every pixel is labelled.

Complexity: linear-time in common grey-level implementations (bucket / ordered queues).

STEP 1 · COMPUTE G ON A TINY 1D IMAGE

Seven pixels, *two dark objects*, one bright peak.

INPUT · f

3	1	2	8	2	1	3
p1	p2	p3	p4	p5	p6	p7

DEFINITION

$$g(p_i) = \max(\text{window}) - \min(\text{window})$$

Window = 3 pixels, mirror padding.

GRADIENT · g(p)

PIXEL	WINDOW	MAX	MIN	G(P)	REGIONAL MIN?
p1	3, 3, 1	3	1	**2**	Yes
p2	3, 1, 2	3	1	**2**	Yes
p3	1, 2, 8	8	1	**7**	—
p4	2, 8, 2	8	2	**6**	Yes
p5	8, 2, 1	8	1	**7**	—
p6	2, 1, 3	3	1	**2**	Yes
p7	1, 3, 3	3	1	**2**	Yes

$g = [\ 2, 2, 7, 6, 7, 2, 2\ ]$

p4 is a minimum in g — gradient measures spread, not height.

STEP 2 · FLOOD LOWEST G FIRST

Two basins *plus* an extra one — over-segmentation in miniature.

STEP	H	PIXEL	DECISION	STATE [P1 … P7]
1	2	p1	No labels → seed B1	[B1, --, --, --, --, --, --]
2	2	p2	Neighbour p1 = B1 → assign B1	[B1, B1, --, --, --, --, --]
3	2	p6	No labels → seed B2	[B1, B1, --, --, --, B2, --]
4	2	p7	Neighbour p6 = B2 → assign B2	[B1, B1, --, --, --, B2, B2]
5	6	p4	No labels → seed B3	[B1, B1, --, B3, --, B2, B2]
6	7	p3	Two basins (B1, B3) → WSHED	[B1, B1, W, B3, --, B2, B2]
7	7	p5	Two basins (B3, B2) → WSHED	[B1, B1, W, B3, W, B2, B2]

RESULT

B1	B1	W	B3	W	B2	B2
p1	p2	p3	p4	p5	p6	p7

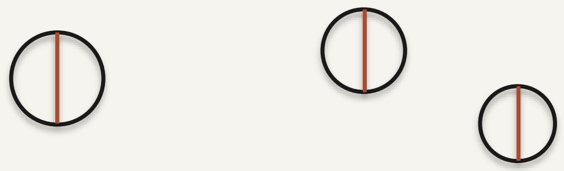
Three basins from two objects — that extra basin around the peak is over-segmentation in miniature. We fix it next.

SAME IDEA IN 2D · SIMPLIFIED SCHEMATIC

Why *g(f)* wins on a real image.

WATERSHED ON F · RAW INTENSITY

Lines cut through bubble interiors.



Dark centres seed basins → red dams cut bubbles in half.

WATERSHED ON G(F) · GRADIENT

Ridges trace bubble contours.

Red contours = watershed lines on g(f). Blue dots = basin seeds.

Schematic only — based on the canonical bubble demo from Beucher & Meyer (1993). Not a real micrograph.

THE CATCH

Every local minimum becomes a *region*.

TYPICAL 256×256 IMAGE

10,000+

basins from noise, texture, and JPEG artefacts. The output is unusable.

WHY

In theory, k regional minima give exactly k basins. Clean.

In practice, **k is huge**. Real images contain thousands of tiny dips in the gradient — from noise, texture, JPEG artefacts — and watershed gives each one its own basin.

THE FIX (NEXT SLIDE)

Tell the algorithm where the regions are. We call these hints markers.

THE FIX

A marker says: *“a region is here.”*

INTERNAL MARKERS

Inside each object

One blob per foreground object. Each seeds that object's basin.

EXTERNAL MARKERS

On the background

Pixels you're sure are not part of any object. Form a single basin that absorbs everything outside.

MARKER IMPOSITION

Geodesic reconstruction

Modify g so its only minima sit at marker locations. Every other minimum is lifted out of the way.

```
Regional_Minima(g*) = M_fg ∪ M_bg
```

MARKER-CONTROLLED WATERSHED IN 5 STEPS

Putting it all together.

STEP 1

Compute g(f)

Morphological gradient of the input.

→

STEP 2

Internal markers

One inside each object of interest.

→

STEP 3

External markers

On the background.

→

STEP 4

Impose to g*

Lift spurious minima via geodesic reconstruction.

→

STEP 5

Watershed on g*

Clean, marker-controlled segmentation.

Pipeline above is a simplified schematic. Real CT not shown.

THREE WAYS IT GOES WRONG

Where watershed *breaks*.

FAILURE 01

Over-segmentation

Spurious minima from noise or texture each seed a basin. Hundreds of regions where only a few objects exist.

FAILURE 02

Boundary leakage

At low-contrast edges, the gradient ridge is too shallow. Floodwater spills — two objects merge into one.

FAILURE 03

Noise sensitivity

Gradient operators amplify high-frequency noise — small perturbations create new minima everywhere.

REACH FOR SOMETHING ELSE WHEN…

Don't use watershed when...

LOW-CONTRAST EDGES

Gradient ridges are too shallow — boundary leakage merges objects.

NO RELIABLE MARKERS

Without seeds, you over-segment into thousands of basins.

HEAVY NOISE / TEXTURE

Gradient amplifies it — spurious minima everywhere.

ALTERNATIVE I · GRAPH CUTS

Global energy minimisation

Smoothness + data term, minimised globally via max-flow / min-cut. Globally optimal where watershed is greedy. Peng & Liu hybrid: 2.44 s → 0.63 s on 256×256.

ALTERNATIVE II · ACTIVE CONTOURS

Snakes / level sets

Deform a curve to minimise an energy functional. Region-based variants need no precise initialisation. Always produces continuous boundary curves.

WHERE THIS GETS USED TODAY

From CT scans to *self-driving* cars.

MEDICAL IMAGING

Lung lobe segmentation from CT

Kaur & Jindal (2011) — marker-controlled watershed for anatomical structure delineation.

DERMATOLOGY

Dermoscopic lesion boundaries

Gilani et al. (2025) — applied to skin lesions; also source of the reliability audit on the previous slide.

INDUSTRIAL

Real-time particle sizing

adaptive watershed for measurement in dynamic fluid flows.

AUTONOMOUS DRIVING

Instance segmentation backbone

Bai & Urtasun (2017) — Deep Watershed Transform replaces the hand-crafted gradient with a learned CNN energy map.

IF YOU REMEMBER NOTHING ELSE…

What to *take away*.

$$g(f) = (f \oplus B) - (f \ominus B)$$

Apply watershed to g(f), not to f.

KEY IDEAS

Image as 3D terrain · intensity as elevation.

Flood from below; build dams where lakes meet.

Apply to gradient g(f) — boundaries are edges.

Naive watershed over-segments. Use markers.

Geodesic reconstruction imposes $\text{Regional\_Minima}(g^*) = M_{fg} \cup M_{bg}$.

COMMON MISCONCEPTIONS

Watershed has no concept of objects — it follows gradient structure.

More regions ≠ better segmentation.

Distance transform is not watershed — it is used to compute markers for it.

Bright peaks can be regional minima in g.

Marker-free watershed is not clinically reliable.